

Carlos Andrés Montoya Tobón

Senior Java Backend Engineer & Solutions Architect

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CORE STACK Java · Spring Boot · Microservices · AWS · Hexagonal Architecture · Keycloak · PostgreSQL · RabbitMQ

PROFESSIONAL SUMMARY

I'm Carlos Andrés Montoya Tobón, a Senior Java Backend Engineer and Solutions Architect based in Medellín, Colombia. For the last 18+ years I've been designing and building Java-based platforms — from transportation management systems and telecom transactional portals to fintech PaaS and blockchain credentials. My specialty is backend architecture with the Spring ecosystem (Spring Boot), microservices and event-driven systems — distributed messaging with Kafka and RabbitMQ, NoSQL at scale with Cassandra — deployed on AWS, GCP and Azure. As Technical Lead and Solutions Architect I've led distributed teams of up to 45 engineers, driven DevOps culture and agile delivery (certified Scrum Master), and helped organizations modernize their stacks and pipelines.

EXPERIENCE

► Solution Architect & Tech Lead

May 2026 — Present

Soldife · Remote

- Set the platform's architectural standard — hexagonal + DDD across 6 microservices and 9 SDKs, dependencies always pointing to the domain, enforced by ArchUnit on every build (zero architecture drift between PRs).
- Designed an ecosystem of 9 auto-configurable Spring Boot SDKs with contract governance — SemVer, backward compatibility and ADRs — published versioned to GitHub Packages.
- Established cross-cutting patterns: transactional outbox, canonical event envelope, end-to-end W3C tracing and centralized identity at the gateway.
- Documented architecture as code: C4 model (L1–L4) in Structurizr, ADRs and AI-oriented development guidelines.
- Designed CI/CD on GitHub Actions (GitFlow + shift-left) with reusable workflows parameterized by project type (Java, KrakenD, NGINX, contracts) and mandatory quality gates: build → tag-filtered unit/Karate tests → JaCoCo coverage → OWASP scan → ArchUnit → Spotless.
- Built SemVer-tagged base images on GHCR (Temurin 21, GraalVM native-image, KrakenD/Go toolchain) to cut per-job tool installs and speed pipelines; org-level governance via branch/PR rulesets and in-pipeline Slack notifications.

Java · Spring Boot · Hexagonal Architecture · DDD · Microservices · ArchUnit · GitHub Actions · KrakenD · GraalVM · Structurizr

Backend Developer

Jan 2026 — Apr 2026

CODERIO · Remote

- Developed backend REST APIs for ITTI, a Paraguayan bank, on Java, Kotlin and Spring Boot behind a Kong API gateway.
- Improved the performance of the core Personas (customers) API — cut response times and raised throughput, profiled and monitored with New Relic.

Java · Kotlin · Spring Boot · Kong · New Relic · REST APIs · Microservices

► Backend Developer

Sep 2024 — Jan 2026

F2X - Flypass · Medellín, Colombia

- Designed a commissions-calculation component for the F2X memberships product (Java · Spring Boot · hexagonal architecture on AWS · Postgres).
- Designed a billing component for the F2X fuel-services product across the microservice fleet.
- Owned production-incident response and operational support for backend services.

Java · Spring Boot · Microservices · Hexagonal Architecture · AWS · PostgreSQL

Backend Developer

Dec 2023 — Dec 2024

Blerify · Remote

- Built security components with Keycloak and implemented KrakenD as the platform's API gateway.
- Developed a blockchain-based voucher/bond issuance component (Java · Vert.x · Kotlin · blockchain · Postgres on GCP/Kubernetes).

Java · Vert.x · Kotlin · Keycloak · KrakenD · Blockchain · PostgreSQL · GCP · Kubernetes

► Backend Developer

Apr 2022 — Jun 2023

Tul · Medellín, Colombia

- Built warehouse-management modules for the TUL B2B marketplace microservice fleet (Java · Spring Boot · hexagonal architecture on AWS · Postgres).
- Delivered Spring Security components and ran platform-wide code-quality reviews.

Java · Spring Boot · Spring Security · Microservices · Hexagonal Architecture · AWS · PostgreSQL

► Solutions Architect

Aug 2020 — Apr 2022

k4s - Klever for Solutions · Medellín, Colombia

- Architected PaaS solutions on AWS for MutualSER's next-gen health-insurance products (Colombian EPS).
- Designed reusable security and Keycloak/GCP integration frameworks consumed across product teams.
- Mentored developers on Spring, SOLID principles and clean architecture.

Java · Spring Boot · AWS · Keycloak · GCP

Solutions Architect

May 2016 — Apr 2020

Integral Software SAS · Medellín, Colombia

- Solutions Architect designing and evolving backend architectures on the Spring ecosystem over a four-year engagement.
- Designed multiple architectures and systems for the Mayor's Office of Medellín — among them Atenea and Tienda Virtual — applying hexagonal architecture, design patterns and security.
- Built components to migrate legacy systems to new architectures for Universidad de Medellín.
- Delivered the same legacy-to-modern architecture migration for the Colombian Air Force (FAC).

Kotlin · Java · Spring Boot · PostgreSQL · React · Angular · Keycloak · Hexagonal

Solutions Architect

Sep 2015 — May 2016

Software Estratégico · Medellín, Colombia

- Defined the new architectures for Matrix Tech's product line on Spring Boot, RabbitMQ and Cassandra.
- Built an SMS messaging gateway from scratch.
- Led the adoption of a DevOps culture to improve communication and collaboration across IT areas, enabling faster, more efficient delivery.

Java · Spring Boot · RabbitMQ · Cassandra · DevOps

► Technical Lead

Aug 2013 — Sep 2015

Tata Consultancy Services · Medellín, Colombia

- Led a distributed team of 45+ developers building Claro Perú's transactional self-service portal, from design through to its production release.
- Designed and built the Oracle Service Bus integration layer connecting Claro Colombia's back-office systems.
- Built the portal itself on Java 8, Spring and WebSphere Portlets, front end included, against IBM WebSphere.
- Stood up the continuous-integration infrastructure and introduced Scrum, giving a 45-person team a single repeatable delivery cadence.

Java · Spring · WebSphere Portlets · Oracle Service Bus · Jenkins

Lead Developer

Nov 2010 — Aug 2013

PATH S.A. · Medellín, Colombia

- Designed NetPlanet analysis components for the Claro network RCs and vehicle-tracking platforms.
- Built Mail Bomber and SMS-delivery systems in Java; IoT pilots for the city of Medellín.
- Designed AWS and on-premise infrastructure and CI/CD pipelines across the organization.

Java · AWS · IoT · CI/CD

Java Web Developer

Jun 2009 — Jun 2010

Ceiba Software House · Medellín, Colombia

- Built the loan, policy and occupational-risk systems of Suramericana — Colombia's largest insurer — on Java / JEE 5 with Struts and JSF.
- Wrote the Oracle data layer behind those flows: SQL and stored procedures carrying the policy and claims operations.
- Delivered every slice full-stack — Java back end, JavaScript / HTML / CSS front end and Oracle schema — deployed on Tomcat.

Java · JEE 5 · Struts · JSF · Oracle SQL · Tomcat

Java Web Developer

Apr 2007 — Jun 2009

Castor SAS (antes Mercurio) · Medellín, Colombia

- Designed and built the business components of a Transportation Management System that ran the daily operation of 25+ transport companies.
- Built the integration layer that carried TMS operations into each client's accounting system, removing the manual re-keying between both worlds.
- Owned support and maintenance for every company on the platform — production incidents, SQL Server / MySQL data fixes and JSP + JavaScript front-end work.
- Introduced Scrum and extreme-programming practices to a team that until then delivered ad hoc.

Java · Servlets · JSP · JavaScript · SQL Server · MySQL · Tomcat

LIVE SITES

MyPipelines mypipelines.andresmontoyat.co — Reusable CI/CD pipeline library — parameterized GitHub Actions workflows (build, tag-filtered tests, JaCoCo coverage, OWASP scan, ArchUnit, Spotless) shared across JVM, KrakenD and NGINX projects.

MyGuidelines — Engineering Standards myguidelines.andresmontoyat.co — My engineering playbook, codified — the architecture, testing, security and clean-code standards I hold every project to, and the trade-offs behind them. Hexagonal + DDD for JVM microservices, written to be executable by engineers and AI coding agents.

TECHNICAL SKILLS

Backend	Java · Kotlin · Spring Boot · Spring Cloud · Spring Security · Hibernate · Kafka · RabbitMQ · Cassandra · REST / gRPC · Keycloak
Cloud & Infrastructure	AWS · GCP · Azure · Kubernetes · Docker · GKE · KrakenD · Terraform
DevOps & Tools	Jenkins · GitHub Actions · SonarQube · Nexus · Gradle · Maven · PostgreSQL · MySQL · Oracle SQL · Scrum Master
AI & Productivity	Claude Code · GitHub Copilot · JetBrains Junie · Prompt Engineering